

POWERED BY **ZAIZENS** · AN INTEGRATED HMS

PREMIUM · PROFESSIONAL EDITION

Radiology & Imaging

Complete staff guide — payment validation, order alerts, exam workflow, structured reporting, modalities, rooms, and clinical integration.

PRODUCT

ZAIZENS Integrated HMS

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Radiology User Manual

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Introduction

Welcome to the **ZAIZENS Integrated HMS Radiology User Manual**. The Radiology module manages the full imaging pathway — from doctor orders and cashier payment verification through exam scheduling, acquisition, structured reporting, and patient report delivery.

Radiology hub: One worklist at `/radiology` for requests, workflow board, validate codes, templates, and consolidated results.

RAD
SERVICE CODES

3
WORKFLOW COLUMNS

All
CLINICAL CONTEXTS



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1. Radiology Overview & Roles

The **Radiology module** connects clinical imaging orders to verified, billable, and reportable exam results across OPD, IPD, Emergency, and Maternity contexts.



Radiology Technician

Validate RAD codes, perform exams, enter structured reports, manage workflow board



Radiologist / Doctor

Orders imaging studies from consultation, IPD, or ER chart; reviews reports



Cashier

Collects payment; issues RAD service codes on order slips



Imaging Room Staff

Room assignment, patient positioning, modality-specific protocols



Radiology Supervisor

Configure rooms and test groups, authorize corrections, review quality



Patient

Pays at cashier, presents RAD code at imaging desk, receives printed report

Key URLs

Screen	Path
Radiology Portal	/portal/radiology
Radiology worklist	/radiology
Validate RAD code	/radiology/validate
Exam workflow board	/radiology/workflow
Radiology request detail	/radiology/requests/:id
Templates / workbench	/radiology/templates
Order alerts	/radiology/order-alerts
Consolidated results	/radiology/results
Configuration	/radiology/configuration
Exam report	/radiology/report/:id

Permissions: radiology.read view orders & results · radiology.write enter & validate results · service_catalog.radiology.write catalog & configuration

2. End-to-End Imaging Workflow

Doctor orders imaging → Patient pays at Cashier (RAD code issued)
→ Radiology validates code → Exam scheduled / performed
→ Results entered (templates) → Finalized → Report on chart & patient copy



Standard OPD pathway

- 1 Doctor adds imaging study (X-ray, ultrasound, CT, etc.) during consultation.
- 2 Patient pays at Cashier — receives slip with RAD-... service code.
- 3 Radiology technician validates code at `/radiology/validate` .
- 4 Exam appears on workflow board — **Start exam** when patient is ready.
- 5 Technician opens radiology template workbench for structured reporting.
- 6 Mark complete — result finalized on patient chart and worklist.
- 7 Print patient copy from report or worklist action menu.

Order alerts shortcut

Doctor orders also appear on **Order alerts** (`/radiology/order-alerts`). Acknowledge → confirm → opens template with patient prefilled.

Payment gate: Radiology cannot process unpaid orders. System blocks validation until Cashier marks the order line as paid.

3. Radiology Portal

The **Radiology & Imaging Portal** (`/portal/radiology`) is the radiology technician home screen.

KPI strip

- **Total Exams** — all registered imaging studies
- **Pending work** — awaiting acquisition or reporting
- **Completed** — finalized reports
- **Orders Today** — new orders registered today

Quick actions

- **All Results** — full worklist at `/radiology`
- **Order alerts** — new doctor orders needing attention
- Portal tiles — role-specific shortcuts (workflow, validate, configuration)

Recent orders table shows latest patients with exam name, modality, status, and date.

4. Radiology Worklist

The **Radiology worklist** (`/radiology`) is the Odoo-style hub for all imaging requests — unified view of OPD orders, formal requests, and legacy results.

KPI strip

Stat	Meaning
Today (open)	Scheduled today, not yet completed
Awaiting acceptance	Submitted requests pending radiology acceptance
In progress	Exam started or reporting underway

Filters

- **Today / All** — date scope
- **Status** — pending payment, submitted, in progress, done, draft, revision
- **Search** — patient name or request number

Worklist columns

Column	Description
Request	Request number; group badge for multi-exam bundles
Patient	Name
Tests	Exam count and description
Scheduled	Date and time
Room	Assigned imaging room
Prescriber	Ordering doctor
Status	Progress badge (e.g. 2/3 done for groups)
Action	Open, Enter result, Report, Print copy

5. Validate Payment Code

Primary entry when a patient arrives at the imaging desk with a payment slip.

- 1 Open `/radiology/validate` or use validate link from portal/worklist.
- 2 Scan barcode or type the **RAD service code** (e.g. `RAD-8821-K2HM9X4P`).
- 3 System verifies: code belongs to radiology · order line is paid · ticket validity passes.
- 4 Validate detail shows patient, referring doctor, and in-house exam lines.
- 5 Create radiology request or open template for structured entry.

Validation failures

Error	Cause
Payment required	Patient has not paid at Cashier
Code belongs to lab/pharmacy	Wrong department code entered
Expired ticket	Consultation-linked validity expired
Duplicate result	Report already registered for this order line

6. Order Alerts

Order alerts (</radiology/order-alerts>) surface doctor-ordered imaging from consultations, IPD, and Emergency.

- 1 Open Order alerts from portal or sidebar.
- 2 Review **Needs your attention** — unacknowledged orders.
- 3 Each alert shows patient, exam name, context (OPD / IPD / Emergency), ordering doctor.
- 4 **Confirm & open template** — prefills patient and order line.
- 5 Or **Mark as seen** if handled via validate-code flow.

Clinical department alerts strip on portal pages highlights radiology items requiring action.

7. Exam Workflow Board

The **workflow board** (`/radiology/workflow`) provides a Kanban view of exams in three columns:

Pending Awaiting start	In progress Exam underway	Completed Report ready
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Card actions

- **Start exam** — moves to In progress; opens RAD template when service code linked
- **Mark complete** — enter findings and conclusion; finalize report
- **Undo** — move back to pending queue
- **View report** — open finalized report from completed column

Each card shows patient initials, exam name, body part, referring doctor, and appointment date.

For custom exams without linked template code, confirm **Start custom report** to begin narrative reporting.

8. Radiology Requests

Formal multi-exam requests use Odoo-style request documents.

Create request

- 1

Validate RAD service code first (required).
- 2

Select patient, prescribing doctor, and scheduled date/time.
- 3

Choose imaging room and test group (multi-exam bundle) or individual catalog exams.
- 4

Submit — request number assigned (e.g. RAD-REQ-00042).

Request lifecycle

Status	Action
Submitted	Radiology accepts request
Accepted	Exams move to in progress on workflow board
In progress	Perform imaging; enter results per line
Done	All exam lines completed — reports available

Open request detail at `/radiology/requests/:id` for line-by-line progress and room assignment.

9. Templates & Structured Reporting

The **Radiology Templates** workbench (</radiology/templates>) provides structured reporting for X-ray, ultrasound, CT, MRI, and custom exam types.

Opening the workbench

- From validate screen after code verification
- From workflow board — Start exam with linked RAD code
- From order alert confirmation
- Direct URL: `?code=RAD-...&oi=...&lock=1&autoload=1`

Structured entry

- 1 Select exam template matching ordered service (modality, body part).
- 2 Enter findings — narrative and structured fields where configured.
- 3 Select conclusion code (normal, abnormal, follow-up recommended, etc.).
- 4 Attach images or documents if PACS integration or upload enabled.
- 5 Save draft or finalize — status moves to **RESULTS READY**.

Test templates maintained under radiology configuration and service catalog.

10. Results Registry

Consolidated results at </radiology/results> and the main worklist provide searchable access to all imaging reports.

Status badges

AWAITING PAYMENT

IN PROGRESS

RESULTS READY

RETURN FOR CORRECTION

Group requests

Multi-exam bundles show progress (e.g. **2/3 done**) until all lines are completed.

Legacy results

Historical exams registered before request workflow link to workflow board by result ID for continuity.

11. Reports & Printing

View report

`/radiology/report/:id` — formatted report with facility header, patient demographics, exam details, findings, conclusion, and modality information.

Print patient copy

- From worklist action menu → Print patient copy
- From report page → `?print=1`
- Diagnostic handover API for integrated print dialogs

Batch scenarios

When multiple exams share one RAD service code, finalize each line individually; batch print when all lines complete.

Reports use facility branding and premium document theme where configured.

12. Corrections & Recalls

Return for correction

Supervisor with `radiology.write` can recall a finalized report:

- 1 Worklist or registry → **Return for correction**.
- 2 Enter audit reason (required).
- 3 Status → **Return for correction** with revision pending flag.
- 4 Re-open via validate code or templates to edit and resubmit.

Recall for edit by code

When linked to billed orders, recall reopens validate-by-code path to maintain payment integrity — same pattern as Laboratory module.

13. External Imaging Results

Studies performed at external facilities are documented separately:

- Mark order line as **external** on validate screen
- Upload report or images via **External upload** or **Attach document**
- Status shows appropriately on worklist
- Document appears on patient chart as attached diagnostic report

Patient obtains study at referral center; staff uploads PDF or DICOM export when results return.

14. OPD, IPD & Emergency Orders

Context	Order source	Payment
OPD	Consultation orders	Prepay at Cashier before validate
IPD	Inpatient chart orders	Credit tab — settle at discharge or per policy
Emergency	ER chart orders	Retrospective billing after care
Maternity	Obstetric imaging orders	Standard cashier or MAT ticket flow

Order alerts show context: Outpatient · Inpatient · Emergency.

Modalities commonly ordered: X-ray, ultrasound, CT, MRI, fluoroscopy — configured in service catalog and room setup.

15. Cashier Integration

Radiology depends on Cashier for payment verification:

1 Doctor order creates OPD order line with pending payment status.

2 Cashier collects fee and marks line paid.

3 System generates RAD-... service code on receipt.

4 Radiology validate checks paid status and ticket validity.

5 Partial payments: patient returns to cashier until balance zero.

See the **Cashier User Manual** (</docs/cashier-users-manual>) for payment methods, wallets, and code validity.

16. Rooms, Groups & Catalog

Imaging rooms

`/radiology/configuration` — define rooms with code, name, modality (X-ray, US, CT, MRI), and active status.

Test groups

Bundle multiple exams (e.g. trauma series, antenatal panel) for single-request workflow.

Service catalog

Imaging studies and prices in Service Catalog → Radiology. Import price lists via `/catalog/radiology/import-price-list` .

Modalities

Standard modalities: CR/DX (X-ray), US (ultrasound), CT, MR (MRI), RF (fluoroscopy). Room and catalog entries align modality for scheduling and reporting.

Requires `radiology.write` or `service_catalog.radiology.write` .

17. Screen Reference (UI)

Radiology Portal — /portal/radiology

Radiology & Imaging Portal

Validate codes · Perform exams · Enter reports

Total 84

Pending 7

Done 72

Today 5

All Results

Order alerts

Workflow

Validate code — /radiology/validate

Radiology · Validate code

RAD-8821-K2HM9X4P

Jean Nkoulou · Dr. Eboa · Chest X-ray PA + Lateral

✓ Paid · Start exam

Workflow board — /radiology/workflow

Pending (3)

Abdominal US

In progress (2)

Chest X-ray

Completed (5)

Skull X-ray

Live screenshots:

Open /portal/radiology or /radiology while signed in; Print → Save as PDF for training.

18. Troubleshooting

Issue	Solution
Code rejected — payment required	Patient must pay at Cashier; verify order line status
Wrong department code	RAD codes for radiology; LAB and PHA go to respective departments
Cannot start exam	Validate RAD code first; check service code linked to order item
Template not loading	Confirm code, oi, and autoload query params; use order alert context
Request stuck on submitted	Accept request on detail page to begin exams
Group request partial	Complete each exam line; progress shows done/total
Print blank report	Verify status is Results ready; try <code>?print=1</code>
Room not listed	Add or activate room under Configuration
External upload fails	Check file size limits; use PDF or image format
Expired consultation ticket	Patient needs new consultation payment if validity expired

19. Glossary

Term	Definition
RAD code	Service code prefix for radiology order lines (e.g. RAD-8821-...)
Modality	Imaging type: X-ray, ultrasound, CT, MRI, fluoroscopy
Worklist	Unified hub of imaging requests at /radiology
Workflow board	Kanban view: pending → in progress → completed
Validate	Verify payment and open order for processing
Template	Structured form for exam findings and conclusion
Test group	Bundle of exams ordered as one request
Imaging room	Physical location/modality assignment for exams
Order alert	Notification of new doctor-ordered imaging study
Revision pending	Report recalled and awaiting correction
External	Study performed outside; results uploaded